

first name \_\_\_\_\_ last name \_\_\_\_\_ email \_\_\_\_\_  
 Dr. Kim's Fall 2026 MTH 207: HW02 (portion to submit) due Sep 24

1. Find  $\lim_{x \rightarrow 10} (x^2 + 3)$  using ONLY the 11 limit laws (to practice the limit laws)

$$= \lim_{x \rightarrow 10} x^2 + \lim_{x \rightarrow 10} 3 = 10^2 + 3 = 103$$

I was asking to ONLY do the 11 limit laws (to practice)

Here is how to do it in a different way (skipping limit laws)

$$\lim_{x \rightarrow 10} (x^2 + 3) = 10^2 + 3 = 103$$

Incorrect to have  $\lim_{x \rightarrow 10} (10^2 + 3)$  in your work.

We either have  $\lim_{x \rightarrow 10}$  or put in 10 but Not Both  
 NB

Do not write  $\frac{1}{0}$

Do not ignore Absolute value

2. Find  $\lim_{x \rightarrow 9} \frac{1}{|x-9|}$  using any techniques from Section 2.2 and 2.3

$$\lim_{x \rightarrow 9} 1 = 1 \quad \lim_{x \rightarrow 9} |x-9| = |9-9| = 0$$

NB "not both"  $\lim_{x \rightarrow 9} |9-9|$

Have either limit or plug in but not both.

| x    | y                                                           |
|------|-------------------------------------------------------------|
| 8.9  | $\frac{1}{ 8.9-9 } = \frac{1}{ -0.1 } = \frac{1}{0.1} = 10$ |
| 8.99 | 100                                                         |

| x    | y                                                          |
|------|------------------------------------------------------------|
| 9.1  | $\frac{1}{ 9.1-9 } = \frac{1}{ 0.1 } = \frac{1}{0.1} = 10$ |
| 9.01 | 100                                                        |

$$\lim_{x \rightarrow 9} \frac{1}{|x-9|} = \infty$$

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Limit notation or plug in for x, but not both

NB  $\lim_{x \rightarrow 3} (3-3)$   $\lim_{x \rightarrow 3} (3^2-9)$

3. Find  $\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$  using any techniques from Section 2.2 and 2.3

$$\lim_{x \rightarrow 3} (x-3) = 3-3 = 0$$

$$\lim_{x \rightarrow 3} (x^2-9) = 3^2-9 = 0$$

zero, so can't split.

$$= \lim_{x \rightarrow 3} \frac{x-3}{(x+3)(x-3)}$$

$$= \lim_{x \rightarrow 3} \frac{1}{x+3}$$

$$\lim_{x \rightarrow 3} 1 = 1$$

$$\lim_{x \rightarrow 3} (x+3) = 3+3 = 6$$

This  $\neq 0$  so we can split

$$= \frac{\lim_{x \rightarrow 3} 1}{\lim_{x \rightarrow 3} (x+3)}$$

$$= \frac{1}{6}$$

Cannot split before cancelling.  
 Split after cancelling

Do not write  $\frac{0}{0}$

$\lim_{x \rightarrow 3} \frac{x-3}{x^2-9}$

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$$\lim_{x \rightarrow 3} \frac{x-3}{(x+3)(x-3)}$$

$$\lim_{x \rightarrow 3} \frac{1}{x+3} = \frac{1}{6}$$

Must mention function here.

because we can't split when the limit of the bottom is zero.

Do not write  $\frac{0}{0}$

Can't split until after canceling

⚠️  $\lim_{x \rightarrow 100} (100 - 100)$   
 ⚠️  $\lim_{x \rightarrow 100} (10 - \sqrt{100})$  NB  
 Either write limit or plug in  $x=100$  but not both

4. Find  $\lim_{x \rightarrow 100} \frac{100 - x}{10 - \sqrt{x}}$  using any techniques from Section 2.2 and 2.3

$$= \lim_{x \rightarrow 100} \frac{(100 - x)(10 + \sqrt{x})}{(10 - \sqrt{x})(10 + \sqrt{x})} \quad \lim_{x \rightarrow 100} (100 - x) = 100 - 100 = 0 \quad \lim_{x \rightarrow 100} (10 - \sqrt{x}) = 10 - \sqrt{100} = 10 - 10 = 0$$

$$= \lim_{x \rightarrow 100} \frac{(100 - x)(10 + \sqrt{x})}{100 - x}$$

$$= \lim_{x \rightarrow 100} (10 + \sqrt{x}) = 10 + \sqrt{100} = 10 + 10 = 20$$

No limit written here

⚠️  $\lim_{x \rightarrow 100} (10 + \sqrt{100})$

Either write limit or plug in for  $x$  but not both

NB

5. Find  $\lim_{x \rightarrow 0} x^4 \sin(\frac{1}{x^3})$  using any techniques from Section 2.2 and 2.3

$$-1 \leq \sin(\frac{1}{x^3}) \leq 1$$

$$-x^4 \leq x^4 \sin(\frac{1}{x^3}) \leq x^4$$

⚠️  $0 \leq x^4 \sin(\frac{1}{x^3}) \leq 0$

⚠️  $0 \leq 0^4 \sin(\frac{1}{0^3}) \leq 0$

Neither of these is between 0 and 0

must mention by name.

NB  
 ⚠️  $\lim_{x \rightarrow 0} 0^4 \sin(\frac{1}{0^3})$

$\lim_{x \rightarrow 0} (-x^4) = -0^4 = 0$  and  $\lim_{x \rightarrow 0} x^4 = 0^4 = 0$  so by the Squeeze Theorem,  $\lim_{x \rightarrow 0} x^4 \sin(\frac{1}{x^3}) = 0$

$\lim_{x \rightarrow 0} x^4 \sin(\frac{1}{x^3}) \neq 0^4 \sin(\frac{1}{0^3})$   $\lim_{x \rightarrow 0} g(x)$  can become  $g(0)$  only when  $g(x)$  is cts at 0, but  $x^4 \sin(\frac{1}{x^3})$  is not cts at 0.

6. Is  $f(x) = \frac{x^2 - 9}{x - 3}$  continuous at  $x = 4$ ? Justify based on the definition of continuous.

$$f(4) = \frac{4^2 - 9}{4 - 3} = \frac{7}{1} = 7$$

$4^2 \neq 8$

$$\lim_{x \rightarrow 4} \frac{x^2 - 9}{x - 3}$$

$$\lim_{x \rightarrow 4} (x^2 - 9) = 7 \quad \lim_{x \rightarrow 4} (x - 3) = 1$$

$$= \frac{\lim_{x \rightarrow 4} (x^2 - 9)}{\lim_{x \rightarrow 4} (x - 3)} = \frac{7}{1} = 7$$

$\lim_{x \rightarrow 4} f(x) = f(4)$  (Both are 7) so  $f$  is continuous at  $x = 4$ .